Quantitative Investing
Problem Set 8

Ashley Wen (xw564) | Emily Wu (esw63) | Grace Gu (xg325) November 9, 2026

Contents. Part 1 covers questions (a)–(c) and (f): the January and December seasonals in the 75 size and past-return portfolios, what they imply for market efficiency, and confidence intervals for the Sharpe ratio of the strategy from (e).

Part 1: Seasonality in Returns

All regressions use the 75 portfolios sorted on size and past 1-month, 2–12-month and 13–60-month returns. The left-hand side is the portfolio return in excess of the T-bill, in % per month. The sample is 1931:02–2026:06 ($T = 1{,}145$), the months in which all three sorts exist. Four portfolios have a few months coded -99.99, which we drop, so their T is slightly smaller (at least 1,127).

Question (a): the January seasonal

We run $r^e_{i,t} = c_i + J_i \, \mathrm{Jan}_t + \varepsilon_{i,t}$ for each portfolio. The intercept c_i is the mean excess return from February to December, and J_i is the extra January return. Table 1 reports J_i for all 75 portfolios.

Table 1: January premium J_i (% per month) from $r^e_{i,t} = c_i + J_i \, \mathrm{Jan}_t + \varepsilon_{i,t}$, OLS t-statistics in parentheses. Rows are size quintiles, columns are past-return quintiles (Low = past losers, High = past winners). 1931:02–2026:06.

Size	Low	2	3	4	High
Panel A: sorted on past 1-month return					
Small	7.32 (6.5)	4.74 (4.8)	4.15 (4.3)	3.72 (4.1)	3.49 (3.7)
2	3.92 (4.0)	3.05 (3.7)	2.33 (3.0)	2.14 (2.6)	1.32 (1.6)
3	3.02 (3.4)	2.35 (3.1)	1.78 (2.5)	1.44 (2.1)	0.39 (0.5)
4	2.15 (2.6)	1.69 (2.4)	1.21 (1.9)	0.40 (0.6)	−0.10 (−0.1)
Big	1.16 (1.6)	0.54 (0.9)	0.15 (0.3)	−0.11 (−0.2)	−0.52 (−0.8)
Panel B: sorted on past 2–12-month return					
Small	7.45 (6.7)	5.14 (5.3)	4.32 (4.8)	3.56 (3.8)	3.57 (3.9)
2	4.07 (3.9)	3.10 (3.6)	2.38 (3.1)	1.68 (2.2)	1.66 (2.0)
3	2.95 (3.0)	2.10 (2.6)	1.68 (2.3)	1.15 (1.7)	0.91 (1.3)
4	2.23 (2.2)	1.45 (1.9)	0.95 (1.4)	0.39 (0.6)	0.34 (0.5)
Big	1.69 (1.9)	0.82 (1.2)	0.13 (0.2)	−0.32 (−0.6)	−0.55 (−0.9)
Panel C: sorted on past 13–60-month return					
Small	7.42 (6.9)	4.73 (5.1)	3.45 (4.0)	4.24 (4.6)	3.70 (3.7)
2	4.56 (4.6)	2.90 (3.4)	2.08 (2.7)	1.71 (2.1)	1.61 (1.9)
3	3.82 (4.1)	2.17 (2.8)	1.73 (2.4)	1.23 (1.8)	0.99 (1.3)
4	2.67 (3.0)	1.29 (1.7)	1.20 (1.7)	0.77 (1.1)	0.28 (0.4)
Big	2.06 (3.1)	0.72 (1.2)	0.31 (0.5)	−0.15 (−0.3)	−0.34 (−0.6)

- **Is there a strong January seasonal different from other months?** Yes, January is by far the strongest month for most of these portfolios. The January premium is positive and significant ($t > 2$) in 43 of the 75 portfolios, and across all 75 it averages 2.1% on top of a

February–December mean of about 0.8%. In other words, an average portfolio earns almost four times its normal monthly return in January.

- **For which portfolios is this largest or smallest?** The effect is largest for small past losers and disappears for large past winners. In every panel the January premium generally falls as we move from small to big stocks and from losers to winners, so the small–loser corner earns more than 7% extra in January with t-statistics near 7. The big–winner corner has a small negative premium that is not significant. The pattern looks the same whether "losers" are defined over 1, 12 or 60 months, so the January effect is a small-stock and past-loser effect at every horizon.
- **Why might this be?** The most common explanation is tax-loss selling. Investors sell stocks with capital losses in December to cut their tax bill, which pushes down prices of past losers, and then buy back in January when the selling pressure ends. Small stocks are hit hardest because they are illiquid, held mostly by taxable individuals, and expensive to arbitrage. Window dressing adds to this: fund managers dump embarrassing losers before year-end reports and buy riskier small stocks again in January. The intercepts also show that past 12-month losers earn only 0.1–0.3% per month from February to December against 1.0–1.3% for winners, so momentum is a February–December phenomenon that reverses in January. All of these stories point to year-end trading pressure, not a change in risk, as the source of the January return.

Question (b): adding December

We add a December dummy, $r^e_{i,t} = c_i + J_i \,\mathrm{Jan}_t + D_i \,\mathrm{Dec}_t + \varepsilon_{i,t}$, so c_i is now the February–November mean. Table 2 reports D_i for all 75 portfolios, and Table 3 compares the three means in the four corner portfolios.

Table 2: December premium D_i (% per month) from $r^e_{i,t} = c_i + J_i \,\mathrm{Jan}_t + D_i \,\mathrm{Dec}_t + \varepsilon_{i,t}$, OLS t-statistics in parentheses. Layout as in Table 1.

Size	Low	2	3	4	High
Panel A: sorted on past 1-month return					
Small	−0.12 (−0.1)	0.10 (0.1)	−0.06 (−0.1)	−0.01 (0.0)	0.29 (0.3)
2	0.68 (0.7)	0.76 (0.9)	0.74 (0.9)	0.99 (1.2)	1.11 (1.3)
3	1.02 (1.1)	0.69 (0.9)	1.12 (1.6)	1.02 (1.5)	1.03 (1.3)
4	0.94 (1.1)	0.80 (1.2)	1.16 (1.8)	0.94 (1.4)	1.17 (1.6)
Big	0.43 (0.6)	0.52 (0.9)	0.67 (1.2)	0.78 (1.4)	0.60 (0.9)
Panel B: sorted on past 2–12-month return					
Small	−0.65 (−0.6)	−0.32 (−0.3)	−0.24 (−0.3)	0.30 (0.3)	1.00 (1.1)
2	0.16 (0.2)	0.35 (0.4)	0.79 (1.0)	1.39 (1.8)	1.46 (1.8)
3	0.62 (0.6)	0.66 (0.8)	0.68 (0.9)	1.27 (1.9)	1.48 (2.0)
4	0.34 (0.3)	0.61 (0.8)	0.79 (1.2)	1.13 (1.8)	1.60 (2.4)
Big	−0.60 (−0.7)	−0.01 (0.0)	0.52 (0.9)	0.90 (1.6)	1.19 (2.0)
Panel C: sorted on past 13–60-month return					
Small	−0.31 (−0.3)	0.07 (0.1)	0.22 (0.2)	−0.10 (−0.1)	0.07 (0.1)
2	0.58 (0.6)	0.91 (1.1)	0.94 (1.2)	0.75 (0.9)	0.51 (0.6)
3	1.00 (1.1)	0.92 (1.2)	1.10 (1.5)	0.71 (1.0)	0.74 (1.0)
4	0.66 (0.7)	0.80 (1.1)	0.98 (1.4)	1.03 (1.5)	0.87 (1.2)
Big	0.65 (1.0)	0.68 (1.1)	0.54 (0.9)	0.67 (1.1)	0.39 (0.6)

Table 3: Corner portfolios from the January–December regression. c is the February–November mean, J and D are the extra January and December returns (% per month, t in parentheses), and the last column is the t-statistic of $J - D$.

Sort	Portfolio	Feb–Nov c	January J	December D	$t(J-D)$
1-month	Small, low	1.42	7.31 (6.5)	−0.12 (−0.1)	4.9
	Small, high	−0.23	3.51 (3.7)	0.29 (0.3)	2.5
	Big, low	0.70	1.20 (1.6)	0.43 (0.6)	0.8
	Big, high	0.47	−0.46 (−0.7)	0.60 (0.9)	−1.2
2–12-month	Small, low	0.12	7.39 (6.6)	−0.65 (−0.6)	5.3
	Small, high	1.15	3.66 (4.0)	1.00 (1.1)	2.2
	Big, low	0.36	1.64 (1.8)	−0.60 (−0.7)	1.8
	Big, high	0.89	−0.44 (−0.7)	1.19 (2.0)	−2.0
13–60-month	Small, low	0.85	7.39 (6.8)	−0.31 (−0.3)	5.3
	Small, high	0.50	3.70 (3.7)	0.07 (0.1)	2.7
	Big, low	0.57	2.12 (3.2)	0.65 (1.0)	1.6
	Big, high	0.63	−0.31 (−0.5)	0.39 (0.6)	−0.8

- **Is there a strong December and January seasonal, and are they different from each other?** Yes for January, but December is only a modest seasonal, and the two are clearly different for small stocks. Adding the December dummy barely changes the January premia. December premia are mostly positive, averaging 0.6% across the 75 portfolios, but only 2 of 75 have $t > 2$ with OLS standard errors. December is a calmer month than the rest of the year, so with White standard errors 12 portfolios reach $t > 2$, 8 of them 12-month winners. December is therefore a real but modest seasonal, about 1–1.6% in a few portfolios. January exceeds December significantly in 21 portfolios, all but one of them in the two smallest size quintiles, while for big stocks the two months cannot be told apart.
- **For which portfolios are these seasonalities largest or smallest?** The two seasonals run in opposite directions across portfolios. The January effect is largest for small losers, while December is weakest there: small stocks earn nothing significant extra in December, and small 12-month losers are even slightly negative. December is largest for mid-to-large 12-month winners, where it is about 1–1.6% and significant with White standard errors, and these are exactly the portfolios with no significant January effect. So small losers do badly into year-end and then rebound, while large winners do well into year-end and, if anything, give a little back in January.
- **Can you conjecture why these differences exist?** The mirror-image pattern is what year-end selling pressure would produce. In December, tax-loss sellers and window dressers sell losers and hold or buy winners, so losers lag and winners gain. In January the pressure lifts, and the losers bounce back while winners are sold to realize gains in the new tax year. The effect concentrates in small stocks because their prices move most for a given amount of selling. The December premium in mid and large stocks may also reflect the pre-holiday effect and year-end inflows into the broad market, which would explain why mid and large stocks earn a little extra in December across most past-return groups.

Question (c): market efficiency

- **Are these findings evidence for or against market efficiency?** At face value they are evidence against it, but weaker evidence than the t-statistics suggest. A return pattern tied to the calendar is known in advance, so in an efficient market investors should buy small losers in

late December and trade the January premium away. Its strength over 95 years, at t-statistics near 7, is hard to explain with risk. Still, several caveats below mean the findings do not cleanly reject efficiency.

- **How would you try to reconcile them with the efficient markets hypothesis?** There are three ways. First, statistics: we looked at 75 portfolios and could have tested any of 12 months, so some large t-statistics are expected by chance. The effect has also weakened sharply since it was published in the late 1970s and early 1980s: the January premium is significant in 46 of 75 portfolios before 1983 but in only 6 after, which is what we would expect if investors learned about it and traded it away. Second, transaction costs: small losers have wide bid–ask spreads and low prices, so the premium may not survive realistic trading costs and short-sale limits. Third, the model: we assumed expected returns are constant across months, and EMH only says prices reflect information given the right model of risk. If small stocks carry more risk in January, or if taxes make the after-tax return equal across months, then a higher pre-tax January return is just compensation. This is the joint-hypothesis problem: we are always testing efficiency and the risk model together.
- **How are the findings perhaps a violation of efficient markets?** The risk story is hard to defend, because nothing about a firm's risk changes on January 1. The tax story only explains why some investors sell, not why arbitrageurs without tax motives fail to buy. The effect is also very large relative to normal returns and has been documented in many countries, including ones with different tax years (Gultekin and Gultekin, 1983), which makes pure data mining less likely. If the premium survives after costs, it is a violation at least of the weak form, since the signal uses only past prices and the calendar. Either way, it shows that limits to arbitrage let predictable patterns persist in the parts of the market that are hardest to trade.

Question (f): confidence intervals for the Sharpe ratio

We use the monthly excess returns of the frozen strategy from (e), both in the walk-forward training period (1953:03–2001:06, $T = 580$) and in the held-out test period (2001:07–2026:06, $T = 300$). Let $\widehat{SR}$ be the monthly Sharpe ratio, $\hat{\gamma}_3$ the skewness and $\hat{\gamma}_4$ the (raw, not excess) kurtosis of the returns. Under i.i.d. normal returns (Lo, 2002) the Sharpe ratio is approximately normal with

$$\operatorname{Var}\left(\widehat{SR}\right) \approx \frac{1 + \frac{1}{2}\widehat{SR}^2}{T},$$

and for i.i.d. returns with any distribution (Mertens, 2002; Opdyke, 2007) the variance becomes

$$\operatorname{Var}\left(\widehat{SR}\right) \approx \frac{1 - \hat{\gamma}_3\,\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{SR}^2}{T}.$$

The 95% interval is $\widehat{SR} \pm 1.96\,\mathrm{SE}$, computed in monthly units and then multiplied by $\sqrt{12}$. Table 4 reports both intervals.

Table 4: Annualized Sharpe ratios of the strategy from (e) with 95% confidence intervals. "Normal" assumes i.i.d. normal returns; "General" adjusts for skewness and kurtosis. Net returns subtract the trading and borrowing costs used in (e).

Sample	Returns	Skew	Kurtosis	$\widehat{SR}$ (ann.)	Normal 95% CI	General 95% CI
Training, 1953–2001	Gross	−0.03	11.0	1.53	[1.24, 1.83]	[1.19, 1.88]
($T = 580$)	Net	−0.05	11.1	1.34	[1.04, 1.63]	[1.00, 1.67]
Test, 2001–2026	Gross	0.02	5.8	0.34	[−0.05, 0.74]	[−0.05, 0.74]
($T = 300$)	Net	0.04	5.8	0.11	[−0.28, 0.50]	[−0.28, 0.50]

- **The training Sharpe ratio is clearly positive under both distributions.** The net training Sharpe ratio of 1.34 has a 95% interval of roughly 1.0 to 1.7, so even its lower end is far above zero and above the market's long-run Sharpe ratio of about 0.4–0.5. Taken at face value, this would say the strategy is a very strong one.
- **The general interval is wider, and the gap grows with the Sharpe ratio.** The strategy's returns are nearly symmetric but have fat tails, with kurtosis of about 11 in training against 3 for a normal distribution. Fat tails make the estimated Sharpe ratio noisier, so the general interval is about 15% wider than the normal one in training. In the test period the adjustment barely matters, because the kurtosis term is multiplied by $\widehat{SR}^2$, which is close to zero there.
- **The test interval includes zero, so the training interval is too optimistic.** The held-out net Sharpe ratio of 0.11 falls far below the lower end of the training interval and below the market's 0.58 over the same years, and its own interval runs from about −0.3 to 0.5. Both intervals assume the strategy was fixed before we saw the data, but the signals in (e) were chosen using facts from (a), (b) and (d) over the full sample, and the rule was revised once after earlier test results were seen. Autocorrelation in the returns is small (about 0.07 at one lag), so it does not explain the gap. The training interval therefore measures sampling noise only, not the selection bias that part (g) is about, which is why the strategy looks much weaker once it is run on data it has not seen.

References

Gultekin, M. N., and N. B. Gultekin (1983). Stock market seasonality: International evidence. *Journal of Financial Economics* 12(4), 469–481.

Lo, A. W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal* 58(4), 36–52.

Mertens, E. (2002). Comments on variance of the IID estimator in Lo (2002). Working paper, University of Basel.

Opdyke, J. D. (2007). Comparing Sharpe ratios: So where are the p-values? *Journal of Asset Management* 8(5), 308–336.